

HAProxy Round-Robin Setup Guide

Step-by-Step Deployment Guide with 50/50 Round-Robin Load Balancing

Step 1: Install HAProxy

```
sudo dnf install epel-release -y
sudo dnf install haproxy -y
```

Step 2: Configure Web Servers Internal Ports

Nginx (Portfolio Site) — Edit `/etc/nginx/conf.d/default.conf` and set port to **8080**:

```
server {
    listen 8080;
    server_name portfolio_site.com;
    root /var/www/html/portfolio;
    index index.html;
}
```

Restart Nginx:

```
sudo systemctl restart nginx
```

Apache (Web Site) — Edit `/etc/httpd/conf.d/site-port81.conf` and set port to **81**:

```
Listen 81
<VirtualHost *:81>
    ServerName web_site.com
    DocumentRoot /var/www/html/web
    <Directory "/var/www/html/web">
        DirectoryIndex index.html web.html
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

Restart Apache:

```
sudo systemctl restart httpd
```

Step 3: Configure HAProxy (Round-Robin Mode)

Overwrite `/etc/haproxy/haproxy.cfg` with:

```

global
    log            127.0.0.1 local2
    chroot         /var/lib/haproxy
    pidfile        /var/run/haproxy.pid
    maxconn        4000
    user           haproxy
    group          haproxy
    daemon
    stats socket   /var/lib/haproxy/stats

defaults
    mode                http
    log                 global
    option              httplog
    option              dontlognull
    option http-server-close
    option forwardfor   except 127.0.0.0/8
    option              redispatch
    retries             3
    timeout http-request 10s
    timeout queue       1m
    timeout connect     10s
    timeout client      1m
    timeout server      1m
    timeout http-keep-alive 10s
    timeout check       10s
    maxconn             3000

frontend http_front
    bind *:80
    mode http
    default_backend web_cluster_backend

backend web_cluster_backend
    mode http
    balance roundrobin
    server nginx_server 127.0.0.1:8080 check
    server apache_server 127.0.0.1:81 check

listen stats
    bind *:8404
    mode http
    stats enable
    stats uri /
    stats refresh 5s

```

Step 4: Configure CentOS Permissions & Firewall

```

# Allow HAProxy network connections in SELinux
sudo setsebool -P haproxy_connect_any 1

# Allow traffic in Firewallld
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --permanent --add-port=8404/tcp
sudo firewall-cmd --reload

```

Step 5: Test & Start HAProxy

```
# Check syntax
sudo haproxy -c -f /etc/haproxy/haproxy.cfg

# Enable and start service
sudo systemctl enable --now haproxy
```

Step 6: Verify Round-Robin Load Balancing

Run these commands multiple times to see requests alternate sequentially between Nginx and Apache:

```
curl -i http://localhost
curl -i http://localhost
```